

Runtime Stuff as a function of input size

Given n integers in the range $[0, N - 1]$

$O(n + N)$ time, # bits to represent the input

Since input is in $[0, N - 1]$, that means the input size is actually $\Theta(\log N)$ so runtime is actually $\Theta(n \log N)$

Eg $n = O(i)$ runtime is now $O(N)$ and input size is $O(\log N)$

Thus if we think back to the arc conflicting set if we have a sufficiently large n -arc we could significantly inflate the problem.